Tile Selection Helps Perception but Hurts Spatial Reasoning in Ultra-High-Resolution Remote Sensing Visual Question Answering

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Abstract. Vision-language models (VLMs) must downscale remote sensing scenes of ultra-high resolution (UHR), so a common remedy is to select tiles relevant to the question and reason over them at native resolution. We asked whether this loses the spatial relations between objects in different tiles, and whether restoring them explicitly helps. We built a hierarchical pipeline that registers objects from selected tiles in a global coordinate frame and computes a geometric relation graph, and compared it with three baselines (whole-image input, independent tiles, and tile selection without relations) on 3,080 XLRS-Bench items using Qwen2.5-VL-7B. The comparison is paired, and the key contrast holds VLM calls and encoded pixels fixed. The explicit relations did not help: the change on relational tasks was +0.0 pp (95% CI [−2.1, +2.1]). The experiment instead shows a significant crossover. Tile selection raises accuracy on perception questions by +4.1 pp but lowers it on relational questions by −4.5 pp (interaction −8.6 pp, $p < 10^{-4}$), which puts object spatial relationships near chance. We trace the null result to the step before the relations: the VLM used as an object extractor returned no objects for 66% of scenes. The results point to routing between global and local views, and to more reliable grounding, rather than to richer relation encodings.

Keywords: Remote sensing · Vision-language models · Ultra-high resolution · Spatial reasoning

1 Introduction

UHR remote sensing images are 8,500×8,500 pixels or larger. XLRS-Bench [19] made plain what this means for VLMs: a target object may cover a few hundred pixels, and whole-image encoding [1] downscales it away. The field's answer is to *zoom*, whether by searching [21,16], pruning tokens [11], retrieving regions [24], or learning an active crop policy [10], each trading global context for local detail.

We tested a specific concern with that trade. When evidence is spread across tiles, the model sees each crop in isolation and may lose the spatial relations between them. We therefore built an explicit remedy (Fig. 1). Objects extracted from the selected tiles are lifted into one global pixel frame and deduplicated across tile seams. A geometric relation graph (direction, distance, adjacency,

containment, intersection) is then passed to the final VLM call. Before running the experiment we fixed the confirmatory hypothesis: the relation block would improve accuracy on relational sub-tasks and matter little on the others.

The hypothesis was not supported. The controlled design nonetheless yields three contributions:

- a **compute-accounted comparison of four methods**, in which *Selection* and *Proposed* run one code path differing by a single flag, with identical VLM calls and pixels;
- a **crossover**: question-conditioned zoom changes accuracy by +4.1 pp on perception and −4.5 pp on relational questions, and independent tiling hurts both;
- a **tightly bounded null** for explicit relations (±2 pp), traced to grounding: extraction returned no objects for 66% of scenes.

2 Related Work

High-resolution VLMs. Tiling encoders such as Monkey [8] and LLaVA-UHD [22], and native dynamic-resolution models such as Qwen2.5-VL [1], raise the input pixel budget. Training-free methods add search or hierarchy at inference time: V* [21], ZoomEye [16], and DC^2 [20]. **UHR remote sensing.** XLRS-Bench [19] and GeoLLaVA-8K [18] target 8K-scale scenes. LRS-VQA [11] prunes tokens under text guidance, ImageRAG [24] retrieves relevant regions, and ZoomEarth [10] learns an active zoom policy. UHR-Micro [13] documents a "resolution illusion" for small evidence; GeoChat [7] is a domain-tuned VLM. **Spatial reasoning and structured intermediates.** VLMs are weak at spatial relations [2,15]. Scene graphs [6], visual programs [4,17], and visual marks [23] externalise that structure; we test it for cross-tile relations under matched compute.

3 Method

Selection. Each scene is cut into 1024-pixel tiles with 128 pixels of overlap (93 tiles per scene on average). Tiles are scored by $0.35\, s_{\text{sal}} + 0.65\, s_{\text{clip}}$. Here s_{sal} is gradient and variance energy on a downscaled overview, and s_{clip} is the CLIP [14] similarity between the question and each tile thumbnail. The top-5 tiles are kept, suppressing near-duplicate neighbours. **Local extraction and registration.** For each selected crop, the VLM returns JSON-formatted objects with boxes on a 0–1000 grid over the crop, which we lift to global pixels in code. Detections are then merged across tiles by a label-aware greedy rule (IoU, containment, or near-touching boxes at a seam), which yields at most 60 objects per scene. **Relations.** For each object and its 6 nearest neighbours, we emit one predicate per pair, in priority order: containment (intersection over the smaller box $\geq$0.85); intersection (IoU $\geq$0.05 or overlap $\geq$5% of the smaller box, e.g. roads crossing rivers); adjacency (border gap $\leq$1.2% of the image diagonal); otherwise 8-way

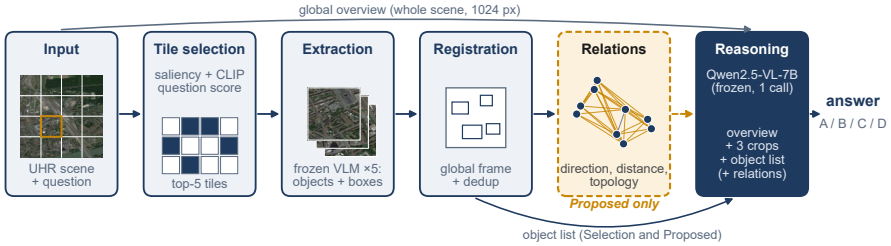


Fig. 1. Architecture of the four compared methods. *Selection* and *Proposed* execute the same code and differ only in whether the relation block (dashed) reaches the final prompt, so both issue 5+1 VLM calls over the same pixels. *Direct* sends the whole image in one call; *Tiling* answers on 8 grid tiles independently and reconciles them by a text-only vote. Real input and graph examples are in Appendix A.3.

compass direction with pixel distance. At most 220 relations are kept, with cross-tile edges ranked first. **Final call.** The final call receives the overview, 3 crops, and the object list in global 0–1000 coordinates. In *Proposed* it also receives the relation lines. The answer-format text is identical across methods.

4 Experimental Setup

Data. We use the XLRS-Bench-lite split [19]: 3,080 four-option multiple-choice items (chance level 25%) in 13 categories. 97 of the 100 Overall-Land-Use items are multi-select (e.g. answer "BCD") and were excluded from every method, leaving 2,983 scored items. Before the experiment we fixed the relational group: object spatial relationship, regional counting, counting with complex reasoning, and route planning (800 items). The remaining 2,123 items form the non-relational group. A 60-item bi-temporal category not matched by our taxonomy is held out of both groups but counted overall. **Model and budget.** We used Qwen2.5-VL-7B-Instruct [1] in bf16 with greedy decoding and a budget of 1,280 visual tokens per image, on one NVIDIA L40S (35.6 GPU-hours in total). **Statistics.** All comparisons are paired on items. We use McNemar's exact test [12] and 10,000-sample paired bootstrap confidence intervals [3]. Per-task tests are Holm-corrected [5]. The two group-level *Selection* vs *Proposed* tests are confirmatory, and all other analyses are exploratory.

5 Results

5.1 The relation block does not help

Table 1 gives the confirmatory result. On relational tasks, *Proposed* and *Selection* score the same (33.4% each). The difference is +0.0 pp (95% CI [−2.1, +2.1]; 37

Table 1. Accuracy (%) and cost per item. Figure 2a gives the 95% CIs. MPix counts the pixels actually encoded by the vision tower. *Selection* and *Proposed* match exactly in calls and pixels, and differ only in prompt tokens.

Method	Calls	MPix	Tokens	s/item	Overall	Relational	Non-rel.
Direct	1.00	0.98	1,403	3.0	41.3	**37.9**	42.6
Tiling	9.15	7.83	11,810	5.0	37.8	34.0	39.0
Selection	6.10	7.53	12,006	16.8	**43.1**	33.4	**46.7**
Proposed	6.10	7.53	12,455	16.9	42.7	33.4	46.2

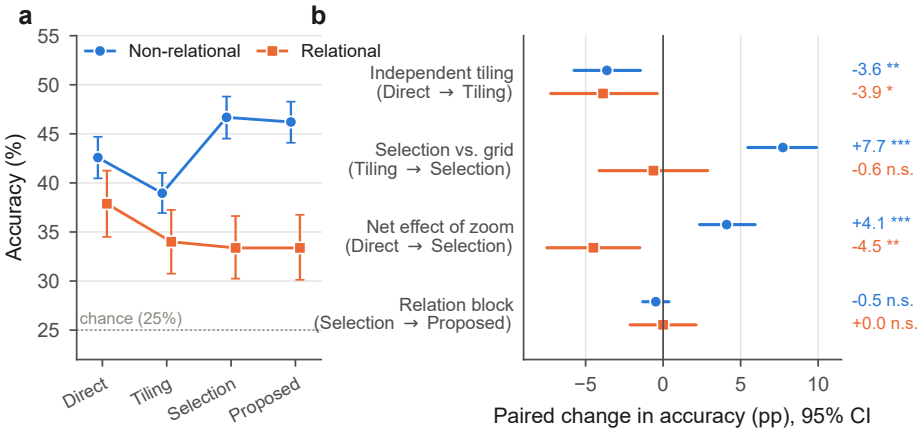


Fig. 2. (a) Accuracy by method and task group, with 95% bootstrap CIs. (b) Paired changes in accuracy with 95% CIs; stars mark McNemar exact p (*<.05, **<.01, ***<.001). The effects of zoom have opposite signs for the two task groups, and the relation block has no effect.

wins against 37 losses; $p = 1.00$). On non-relational tasks the change is -0.5 pp $[-1.3, +0.4]$ ($p = 0.31$). The two methods match in calls (6.10) and encoded pixels (7.53 MP); the relation block adds 448 prompt tokens on average and has no measurable benefit. The interval excludes gains above about 2 pp.

5.2 Zoom helps perception and hurts relations

The progression across methods (Fig. 2) is more revealing. *Tiling* is worse than *Direct* in both groups (-3.6 pp, $p = 2 \times 10^{-4}$) despite 9× the calls. *Selection* recovers $+7.7$ pp over *Tiling* on non-relational tasks but nothing on relational ones (-0.6 pp). Relative to *Direct*, zooming therefore changes sign with the task group: $+4.1$ pp $[+2.4, +5.9]$ for non-relational against -4.5 pp $[-7.5, -1.5]$ for relational tasks, a difference of -8.6 pp $[-12.1, -5.1]$ (permutation $p < 10^{-4}$). On object spatial relationship (500 items), accuracy falls from 32.4% with *Direct* to 27.2% with *Selection*, near the 25% chance level. Per-task effects (Fig. 3) agree in

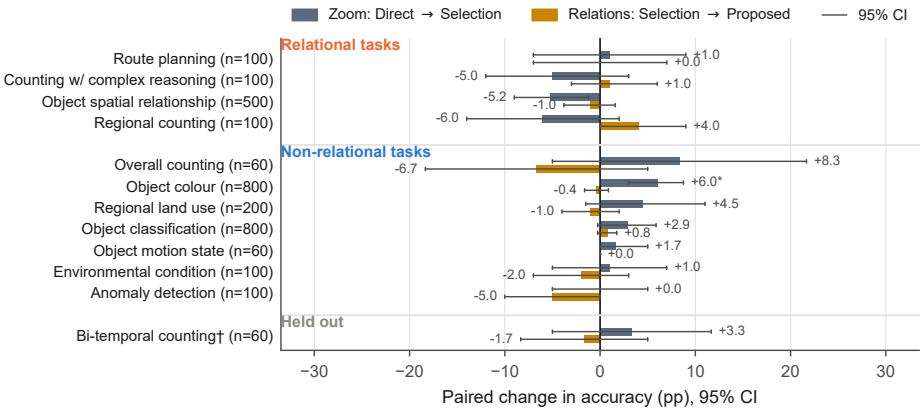


Fig. 3. Paired per-task effects with 95% CIs. Slate bars show the effect of zooming (*Direct* → *Selection*), amber bars the effect of the relation block (*Selection* → *Proposed*). * marks Holm-adjusted $p < .05$. Overall land use (n=3 scorable) is omitted. †The bi-temporal category is held out of both groups.

direction, though only object colour survives Holm correction. This exploratory result fits our premise that zooming discards the global layout relational questions need; re-supplying it as text did not repair the loss.

5.3 Why explicit relations did not help

The relation graph was usually empty. On 66% of scenes, the VLM extractor returned no parseable object in any of the five selected crops (Fig. 4a). For object colour, classification and motion state, the rate was 98–100%. Only 21% of scenes produced any cross-tile relation. When objects were found, graphs were rich (16.3 objects and 59.3 relations on average), and parsing was not the cause (Appendix A.4). **When relations were present, the model used them, and the effect depended on the task group.** We use the rate at which the answer changes between *Selection* and *Proposed* as a probe. It is 5.6% when the relation block is empty; this is the noise floor from the prompt text alone. It rises to 17.4% when relations are present, and to 30.0% on relational items with cross-tile edges (Fig. 4b). The sign of the effect followed our hypothesis in direction but was not significant: +2.8 pp [−2.0, +7.7] on relational items with cross-tile relations ($n = 247$), and −4.1 pp [−7.5, −1.0] on non-relational ones ($n = 293$; uncorrected $p = 0.02$). On perception questions the relations therefore act as distractors (post-treatment strata; indicative only). The rise of accuracy with object count is not evidence for the method: it appears in *Direct* as well, which never sees any objects (Appendix, Fig. 7). Object-rich scenes simply belong to easier tasks.

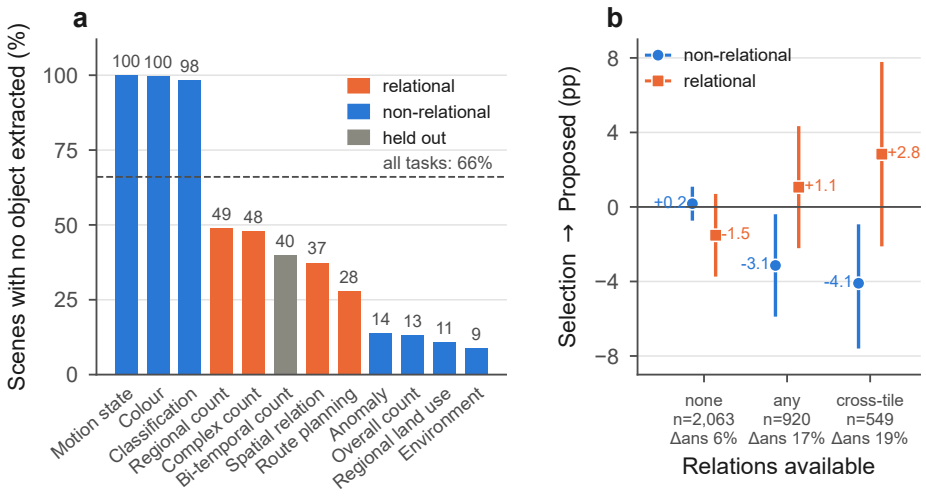


Fig. 4. (a) Share of scenes where the VLM extractor returned no object in any of the five selected crops. (b) Change from *Selection* to *Proposed*, stratified by what the relation block contained. Δans is the share of items whose answer changed; 5.6% with an empty block is the noise floor from the prompt text alone.

5.4 Complementarity and cost

Direct and *Selection* fail on different items. An oracle that picks the correct one of the two reaches 51.2% overall (44.9% relational, 53.6% non-relational), against 43.1% for the best single method. *Selection* and *Proposed* cost 5.6× the wall time of *Direct*, and local extraction accounts for 77% of that time, while their net gain is +1.8 pp. A router that decides *whether* to zoom could capture most of the complementarity at a fraction of the cost.

6 Discussion and Conclusion

Under matched compute, explicit cross-tile relations did not improve UHR remote sensing VQA. Two lessons follow. First, zoom-based pipelines trade relational accuracy for perceptual accuracy, and aggregate scores hide the trade, so relational and perception tasks should be reported separately. Second, a symbolic relation layer is only as good as its grounding: a 7B VLM prompted as a detector returned nothing on most scenes, precisely on the small-object questions that motivate zooming. A dedicated detector is the necessary next test. **Limitations:** one 7B model, the multiple-choice lite split only, a generic CLIP scorer [9], and a relational partition of our own design (Appendix A.2).

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A Supplementary Material

A.1 Per-task results

Table 2. Per-task accuracy (%) and paired effects. Δ_z is the change from *Direct* to *Selection*; Δ_r is the change from *Selection* to *Proposed*. Grp is R for relational tasks, N for non-relational, and H for the held-out category. The last column gives the share of scenes where the extractor returned no objects.

Task	Grp	n	Direct	Tiling	Select.	Prop.	Δ_z	Δ_r	No obj.
Complex counting	R	100	56.0	43.0	51.0	52.0	−5.0	+1.0	48
Object spatial rel.	R	500	32.4	30.2	27.2	26.2	−5.2	−1.0	37
Regional counting	R	100	45.0	36.0	39.0	43.0	−6.0	+4.0	49
Route planning	R	100	40.0	42.0	41.0	41.0	+1.0	+0.0	28
Anomaly detection	N	100	70.0	65.0	70.0	65.0	+0.0	−5.0	14
Environ. condition	N	100	79.0	72.0	80.0	78.0	+1.0	−2.0	9
Object class.	N	800	35.0	31.5	37.9	38.6	+2.9	+0.8	98
Object colour	N	800	35.4	34.4	41.4	41.0	+6.0*	−0.4	100
Motion state	N	60	65.0	65.0	66.7	66.7	+1.7	+0.0	100
Overall counting	N	60	25.0	16.7	33.3	26.7	+8.3	−6.7	13
Overall land use	N	3	66.7	66.7	66.7	66.7	+0.0	+0.0	0
Regional land use	N	200	68.0	56.0	72.5	71.5	+4.5	−1.0	11
Bi-temporal counting	H	60	43.3	46.7	46.7	45.0	+3.3	−1.7	40

A.2 Further caveats

The planned ablation sweeps over tile size, overlap and top-k were not run, so the configuration reported here is a single operating point. All methods chose option "C" for 37–41% of items, against 27% of ground-truth answers: a position bias shared across methods, which affects absolute accuracy but not the paired comparisons. The 97 multi-select Overall-Land-Use items were excluded identically from every method; set-based scoring of these items is left to future work. The stratified analysis in Section 5.3 conditions on the number of relations, which is a post-treatment variable, and its p-values are uncorrected.

A.3 Example inputs, questions and relation graph

Figure 5 shows one scene as the pipeline sees it, and Fig. 6 the relation graph built for it. Table 3 lists real benchmark questions, verbatim.

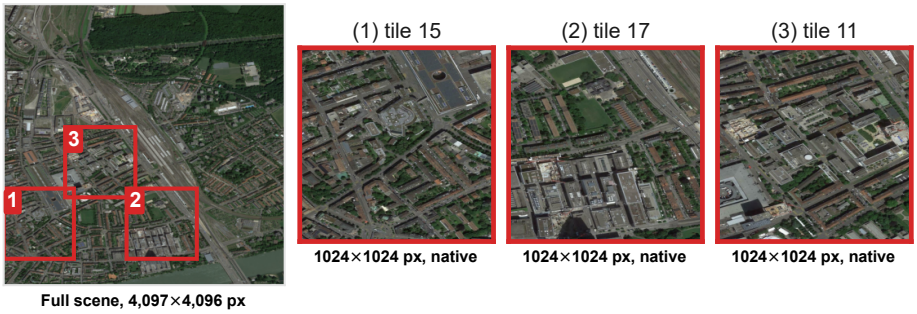


Fig. 5. Input example (object spatial relationship item). Left: the full scene, with the three highest-scoring tiles outlined. Right: the same tiles at native resolution, as the VLM sees them during local extraction.

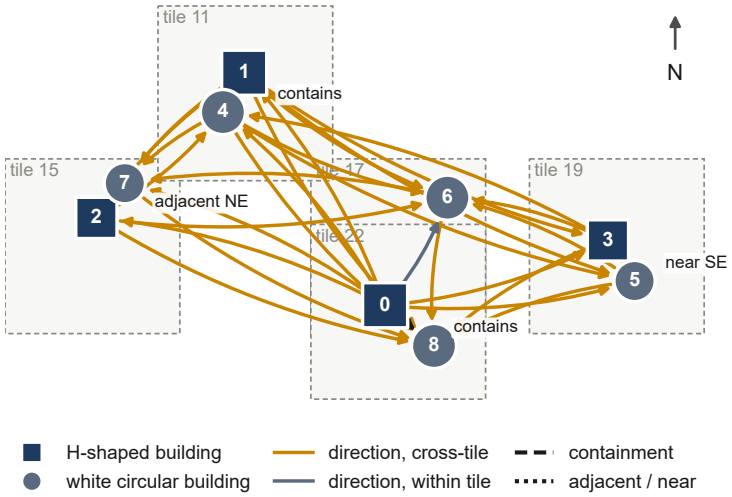


Fig. 6. Relation graph for the scene in Fig. 5. Nodes are placed at their global positions (lightly separated where they overlap); dashed boxes are the selected tiles. 27 of the 32 relations connect objects observed in different tiles. *Proposed* answered this item correctly and *Selection* did not. The labels also show a failure mode: the extractor copied the question's wording into its object labels.

Table 3. Example XLRS-Bench questions (verbatim). R marks tasks in the relational group, N the non-relational group.

Task	Grp	Question (answer)
Counting with Complex Reasoning	R	How many parking lots are there in the town shown in the picture above? *(D: 9)*
Object Spatial Relationship	R	In the image, the two smaller green rectangular greenhouses located towards the upper right are positioned in which direction relative to the larger multi-unit residential area? *(C: To the right)*
Regional Counting	R	How many houses are there inside the red circle? *(B: 5)*
Route Planning	R	In the image below, what is the shortest driving route from the building with the red pyramid roof to the parking lot at the port on the left? *(B: Exit the building and head west. Continue straight through the first intersection. At the next intersection, turn left to reach your destination.)*
Anomaly Detection and Interpretation	N	The width and calmness of the river in the picture suggest? *(D: It may serve as a vital natural water source, crucial for the surrounding ecosystem.)*
Object Classification	N	Identify the object category within the given bounding box in the satellite or aerial image. Image resolution: 4096 x 4097. Bounding box: [2368, 2061, 2447, 2108]. *(B: The Red Roof)*
Object Color	N	Determine the color of the object within the given reference bounding box in the satellite image. Image resolution: 5616 x 3744. Bounding box: [3826, 1648, 3944, 1695]. *(C: White)*
Overall Land Use Classification	N	Select the options that best represent the types of land use depicted in the image. *(ABD: Farmland; Lakes or Ponds; Parking Lot or Garage [multi-select])*
Regional Land Use Classification	N	Select the type of land marked with a red circle in the image. *(D: Farmland)*

A.4 Compute and extraction details

Only 1.6% of the 19,481 items returned by the extractor lacked a readable box, so the empty graphs reflect the extractor returning no objects rather than malformed output. Mean seconds per item for *Proposed*: overview 2.6, selection 0.3, local extraction 13.0, final answer 0.9; deduplication and relation construction together take 0.003. Peak VRAM was 18 GB. Deduplication merged 10.7% of raw detections, 623 of them across tile seams. 22% of the parsed detections touched a tile border.

A.5 Object count is confounded with task

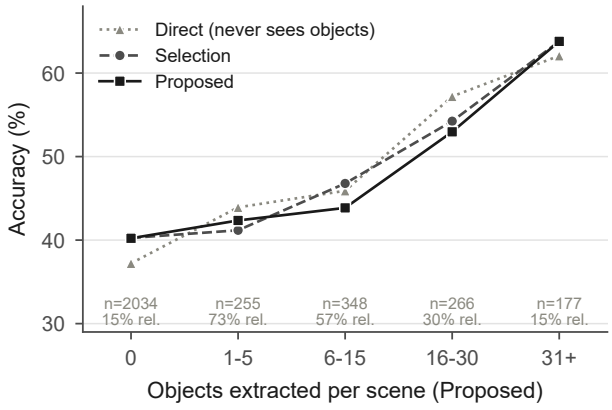


Fig. 7. Accuracy against the number of objects extracted. All methods improve together, including *Direct*, which never receives objects. The trend therefore reflects the mix of scenes and tasks (the labels give the share of relational items per bin) rather than any benefit of the relation graph.

A.6 Qualitative examples

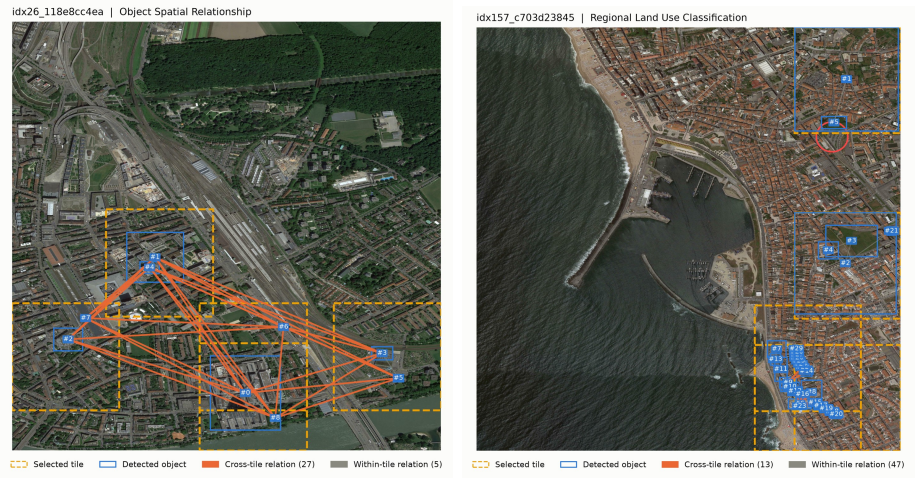


Fig. 8. Left: an object spatial relationship item that *Proposed* answers correctly and *Selection* does not; 27 of its 32 relations span tiles. Right: a regional land-use item where adding 116 relations changes a correct *Selection* answer into an error, an example of relations acting as distractors on a perception task. Dashed boxes are the selected tiles; orange edges are cross-tile relations.